

The Pumping Lemma for Context-Free Languages

Take an **infinite** context-free language



Generates an infinite number
of different strings

Example:

$$S \rightarrow AB$$

$$A \rightarrow aBb$$

$$B \rightarrow Sb$$

$$B \rightarrow b$$

$$S \rightarrow AB$$

$$A \rightarrow aBb$$

$$B \rightarrow Sb$$

$$B \rightarrow b$$

In a derivation of a long string,
variables are repeated

A derivation:

$$S \Rightarrow AB \Rightarrow aBbB \Rightarrow abb\underline{B}$$

$$\Rightarrow abbSb \Rightarrow abbABb \Rightarrow abbaBbBb \Rightarrow$$

$$\Rightarrow abbabb\underline{B}b \Rightarrow abbabbbb$$

Consider now an infinite
context-free language L

Let G be the grammar of $L - \{\lambda\}$

Take G so that it has no unit-productions
no λ -productions

Let $p =$ (Number of productions) $\times$
(Largest right side of a production)

Let $m = p + 1$

Example $G: S \rightarrow AB$

$A \rightarrow aBb$

$B \rightarrow Sb$

$B \rightarrow b$

$$p = 4 \times 3 = 12$$

$$m = p + 1 = 13$$

Take a string $w \in L(G)$
with length $|w| \geq m$

We will show:

in the derivation of w
a variable of G is repeated

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$$S \Rightarrow w$$

$$v_1 \Rightarrow v_2 \Rightarrow \cdots \Rightarrow v_k \Rightarrow w$$

$$S = v_1$$

$$v_1 \Rightarrow v_2 \Rightarrow \cdots \Rightarrow v_k \Rightarrow w$$

$$|v_i| < |v_{i+1}| + f \quad \leftarrow \text{maximum right hand side of any production}$$



$$|w| < k \cdot f$$



$$m \leq |w| \leq k \cdot f \quad \longrightarrow \quad p < k \cdot f$$

$$v_1 \Rightarrow v_2 \Rightarrow \cdots \Rightarrow v_k \Rightarrow w$$

$$p < k \cdot f$$



$$k > \frac{p}{f}$$



Number of productions
in grammar

$$v_1 \Rightarrow v_2 \Rightarrow \cdots \Rightarrow v_k \Rightarrow w$$

$k >$ Number of productions
in grammar



Some production must be repeated

$$v_1 \Rightarrow \cdots \Rightarrow a_1 A a_2 \Rightarrow \cdots \Rightarrow a_3 A a_4 \Rightarrow \cdots \Rightarrow w$$

Repeated
variable

$$\begin{array}{l} S \rightarrow r_1 \\ A \rightarrow r_2 \\ B \rightarrow r_2 \\ \dots \end{array}$$

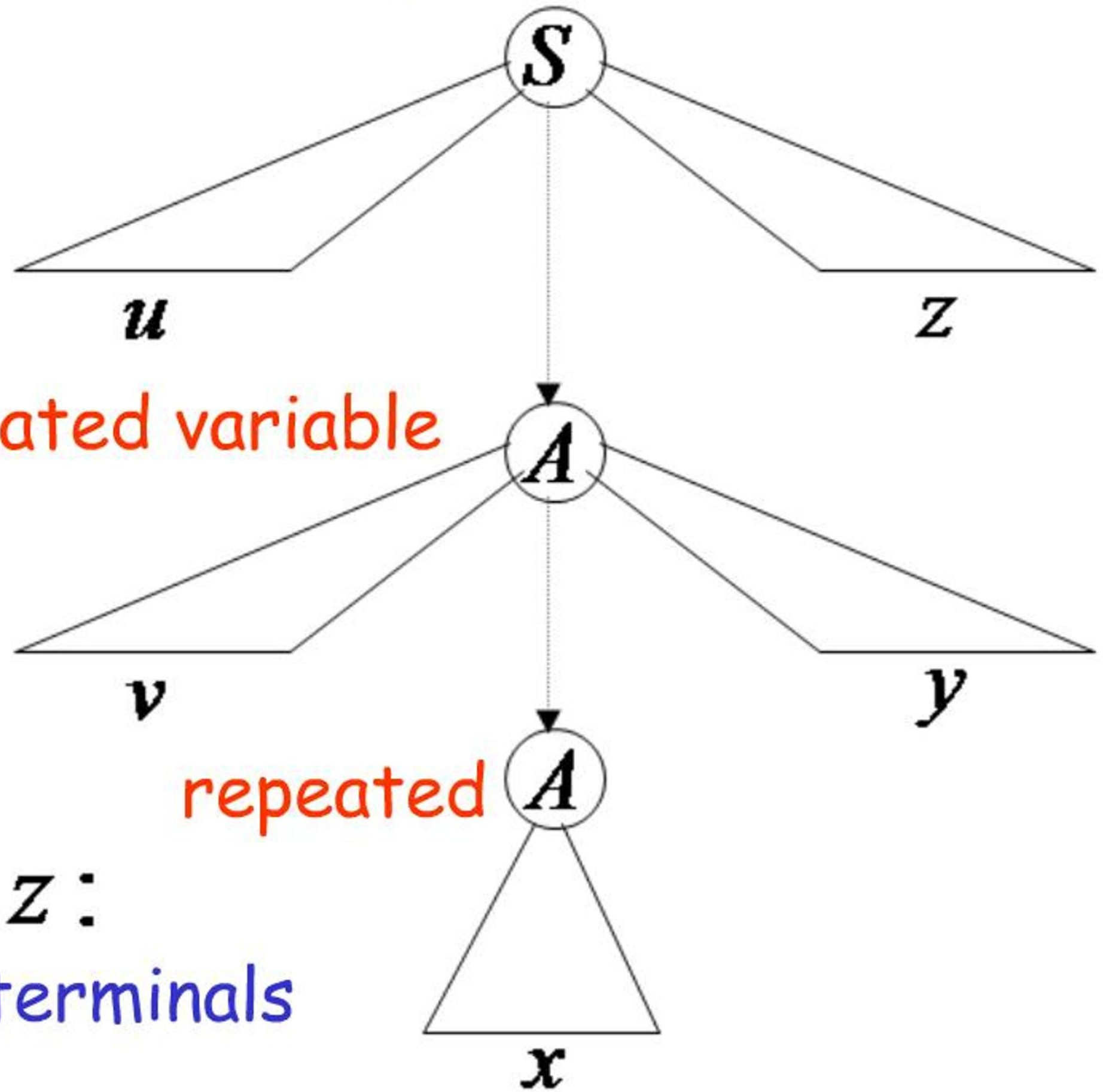
$$w \in L(G) \quad |w| \geq m$$

Derivation of string w

$$S \Rightarrow \cdots \Rightarrow a_1 A a_2 \Rightarrow \cdots \Rightarrow a_3 A a_4 \Rightarrow \cdots \Rightarrow w$$

Some variable is repeated

Derivation tree of string w



Last repeated variable

$$w = uvxyz$$

repeated

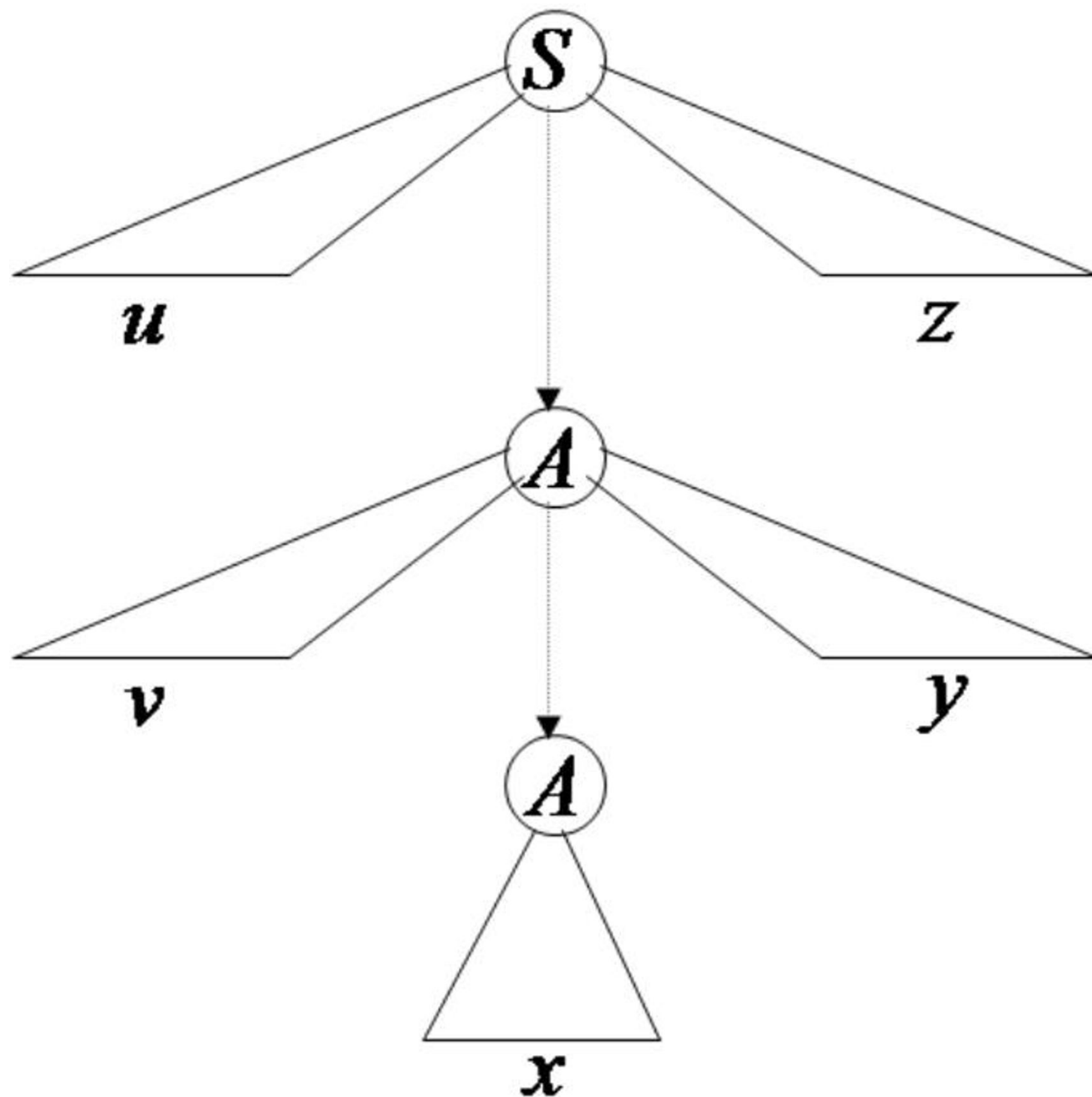
u, v, x, y, z :
Strings of terminals

Possible
derivations:

$$\begin{array}{c} * \\ S \Rightarrow uAz \end{array}$$

$$\begin{array}{c} * \\ A \Rightarrow vAy \end{array}$$

$$\begin{array}{c} * \\ A \Rightarrow x \end{array}$$



We know:

$$\overset{*}{S} \Rightarrow uAz$$

$$\overset{*}{A} \Rightarrow vAy$$

$$\overset{*}{A} \Rightarrow x$$

This string is also generated:

$$\overset{*}{S} \Rightarrow uAz \overset{*}{\Rightarrow} uxz$$

$$uv^0xy^0z$$

We know:

$$\overset{*}{S} \Rightarrow uAz \qquad \overset{*}{A} \Rightarrow vAy \qquad \overset{*}{A} \Rightarrow x$$

This string is also generated:

$$\overset{*}{S} \Rightarrow uAz \overset{*}{\Rightarrow} uvAyz \overset{*}{\Rightarrow} uvxyz$$

The original $w = uv^1xy^1z$

We know:

$$\overset{*}{S} \Rightarrow uAz$$

$$\overset{*}{A} \Rightarrow vAy$$

$$\overset{*}{A} \Rightarrow x$$

This string is also generated:

$$\overset{*}{S} \overset{*}{\Rightarrow} uAz \overset{*}{\Rightarrow} uvAyz \overset{*}{\Rightarrow} uvvAyyz \overset{*}{\Rightarrow} uvvxyyz$$

$$uv^2xy^2z$$

We know:

$$\overset{*}{S} \Rightarrow uAz$$

$$\overset{*}{A} \Rightarrow vAy$$

$$\overset{*}{A} \Rightarrow x$$

This string is also generated:

$$\begin{array}{ccccccc} & * & & * & & * & \\ S \Rightarrow uAz & \Rightarrow & uvAyz & \Rightarrow & uvvAyyz & \Rightarrow & \\ & * & & * & & & \\ \Rightarrow & uvvvAyyyyz & \Rightarrow & uvvvxyyyz & & & \end{array}$$

$$uv^3xy^3z$$

We know:

$$\overset{*}{S} \Rightarrow uAz$$

$$\overset{*}{A} \Rightarrow vAy$$

$$\overset{*}{A} \Rightarrow x$$

This string is also generated:

$$S \xRightarrow{*} uAz \xRightarrow{*} uvAyz \xRightarrow{*} uvvAyyz \xRightarrow{*}$$

$$\xRightarrow{*} uvvvAyyyzyz \xRightarrow{*} \dots$$

$$\xRightarrow{*} uvvv\dots vAy\dots yyyz \xRightarrow{*}$$

$$\xRightarrow{*} uvvv\dots vxy\dots yyyz$$

$$uv^i xy^i z$$

Therefore, any string of the form

$$uv^i xy^i z \quad i \geq 0$$

is generated by the grammar G

Therefore,

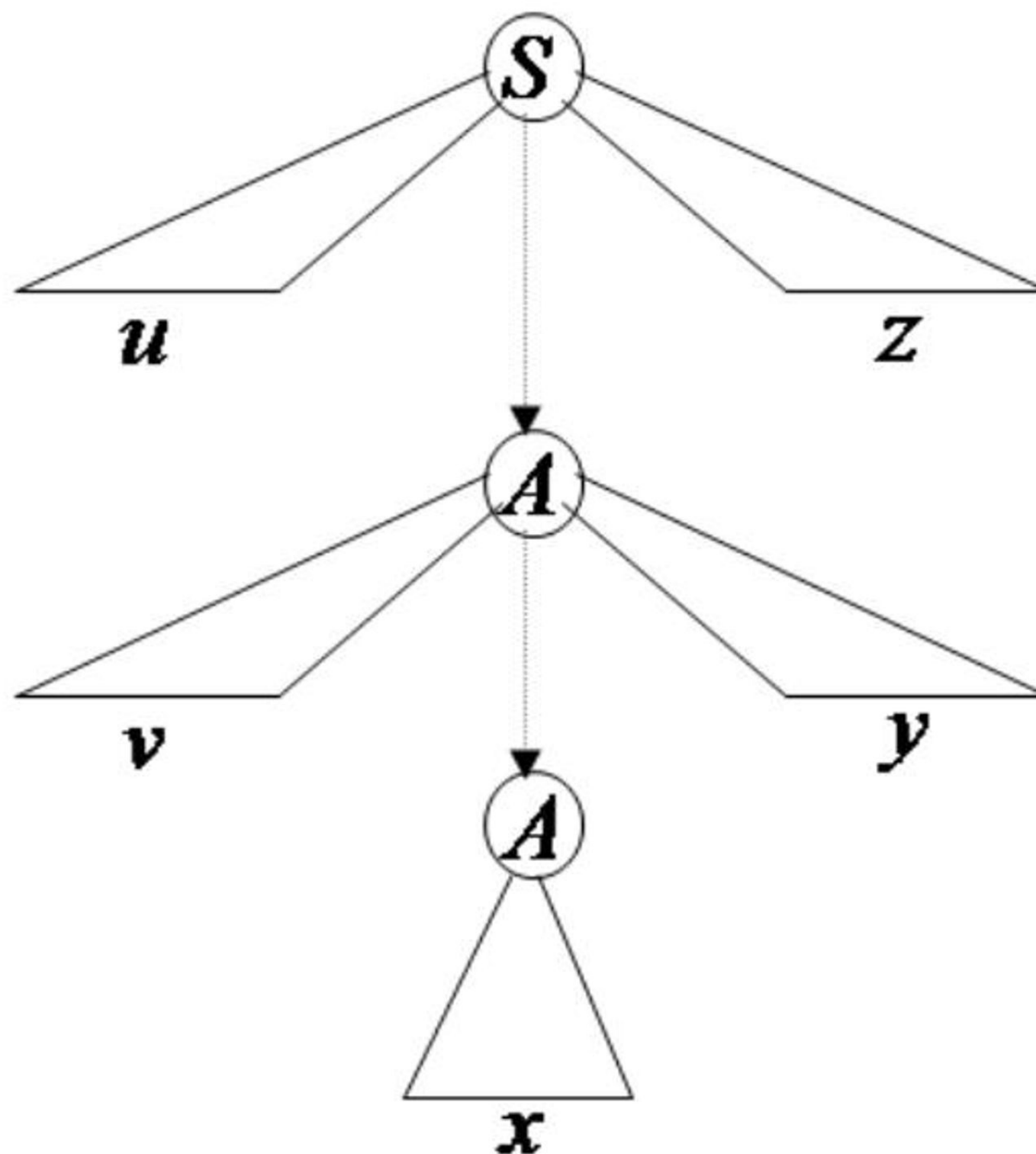
knowing that $uvxyz \in L(G)$

we also know that $uv^i xy^i z \in L(G)$

$$L(G) = L - \{\lambda\}$$

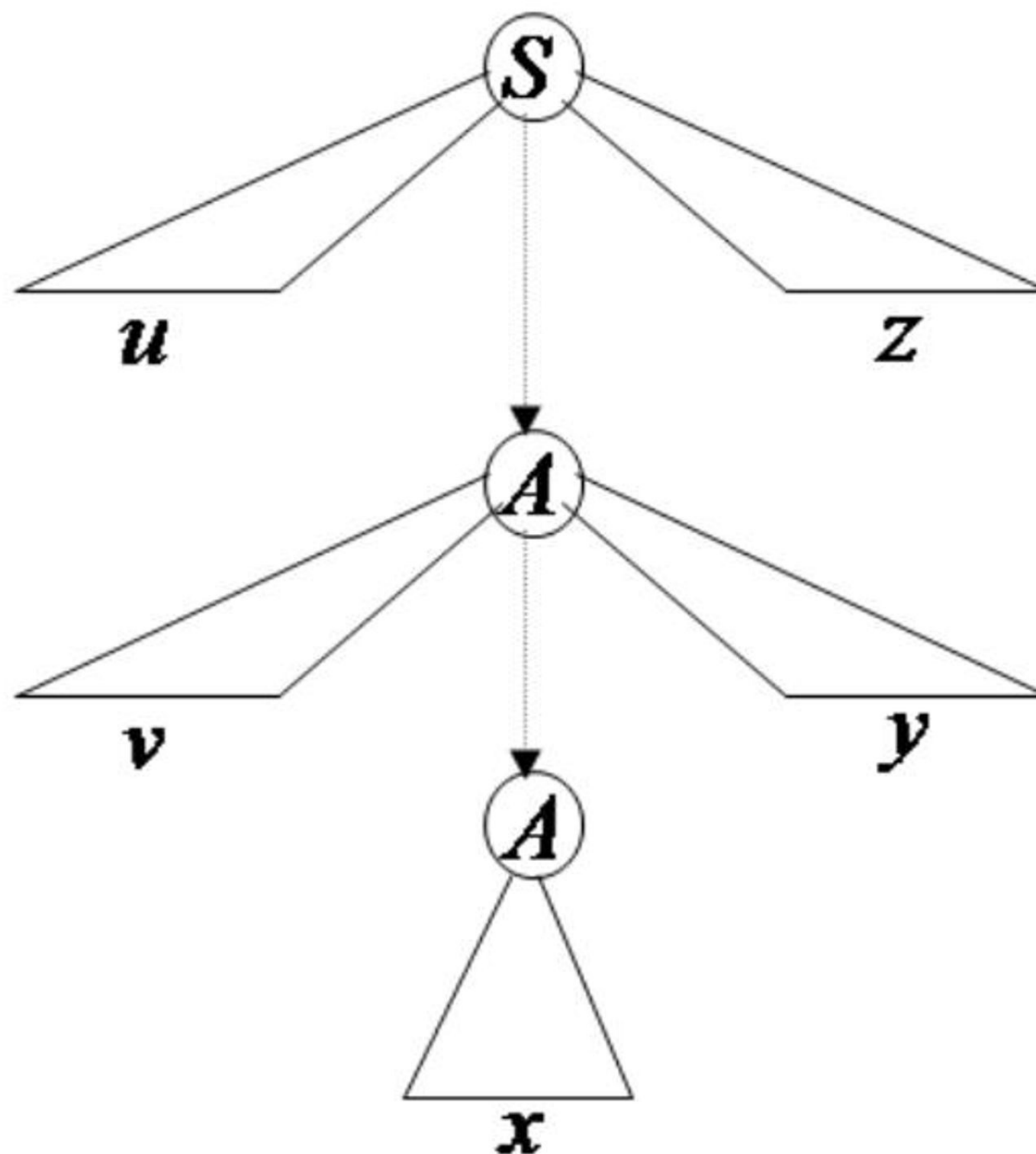


$$uv^i xy^i z \in L$$



Observation: $|vxy| \leq m$

Since **A** is the last repeated variable



Observation: $|vy| \geq 1$

Since there are no unit or λ -productions

The Pumping Lemma:

For infinite context-free language L
there exists an integer m such that

for any string $w \in L, |w| \geq m$

we can write $w = uvxyz$

with lengths $|vxy| \leq m$ and $|vy| \geq 1$

and it must be:

$$uv^i xy^i z \in L, \quad \text{for all } i \geq 0$$

Applications of The Pumping Lemma

Non-context free languages

$$\{a^n b^n c^n : n \geq 0\}$$

Context-free languages

$$\{a^n b^n : n \geq 0\}$$

Theorem: The language

$$L = \{a^n b^n c^n : n \geq 0\}$$

is **not** context free

Proof: Use the Pumping Lemma
for context-free languages

$$L = \{a^n b^n c^n : n \geq 0\}$$

Assume for contradiction that L
is context-free

Since L is context-free and infinite
we can apply the pumping lemma

$$L = \{a^n b^n c^n : n \geq 0\}$$

Pumping Lemma gives a magic number ***m*** such that:

Pick any string $w \in L$ with length $|w| \geq m$

We pick: $w = a^m b^m c^m$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

We can write: $w = uvxyz$

with lengths $|vxy| \leq m$ and $|vy| \geq 1$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Pumping Lemma says:

$$uv^i xy^i z \in L \quad \text{for all } i \geq 0$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

We examine all the possible locations
of string vxy in w

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 1: vxy is within a^m

$$\begin{array}{c} \overbrace{aaa \dots aaa}^m \quad \overbrace{bbb \dots bbb}^m \quad \overbrace{ccc \dots ccc}^m \\ \underbrace{aaa \dots aaa}_{u \quad vxy} \quad \underbrace{bbb \dots bbb}_z \end{array}$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 1: v and y consist from only a

$$\overbrace{aaa \dots aaa}^m \overbrace{bbb \dots bbb}^m \overbrace{ccc \dots ccc}^m$$

$$\underbrace{\quad}_{u} \underbrace{\quad}_{vxy} \underbrace{\quad}_{z}$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 1: Repeating v and y

$$k \geq 1$$

$$\overbrace{aaaaaa \dots aaaaaa}^{m+k} \overbrace{bbb \dots bbb}^m \overbrace{ccc \dots ccc}^m$$

$$\underbrace{\quad}_{u} \quad \underbrace{\quad}_{v^2 xy^2} \quad \underbrace{\quad}_{z}$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 1: From Pumping Lemma: $uv^2xy^2z \in L$

$$k \geq 1$$

$$\overbrace{aaaaaa \dots aaaaaa}^{m+k} \overbrace{bbb \dots bbb}^m \overbrace{ccc \dots ccc}^m$$

$\underbrace{\hspace{1.5cm}}_u \quad \underbrace{\hspace{3.5cm}}_{v^2xy^2} \quad \underbrace{\hspace{3.5cm}}_z$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 1: From Pumping Lemma: $uv^2xy^2z \in L$
 $k \geq 1$

However: $uv^2xy^2z = a^{m+k}b^m c^m \notin L$

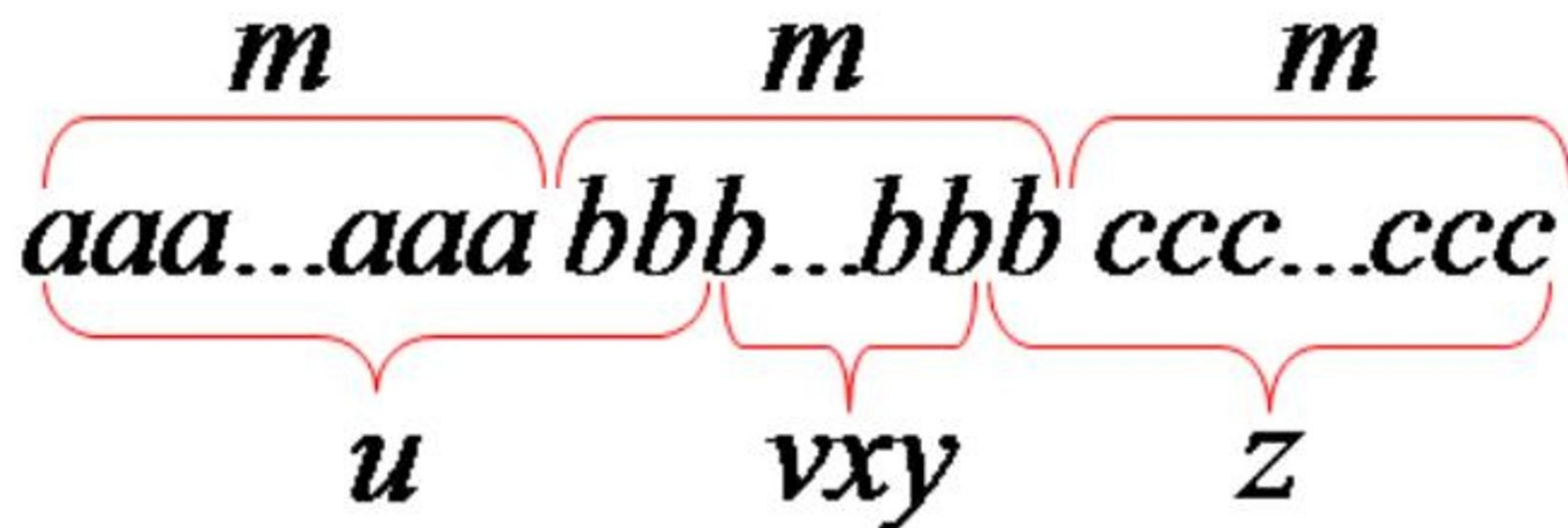
Contradiction!!!

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 2: vxy is within b^m



$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 2: Similar analysis with case 1

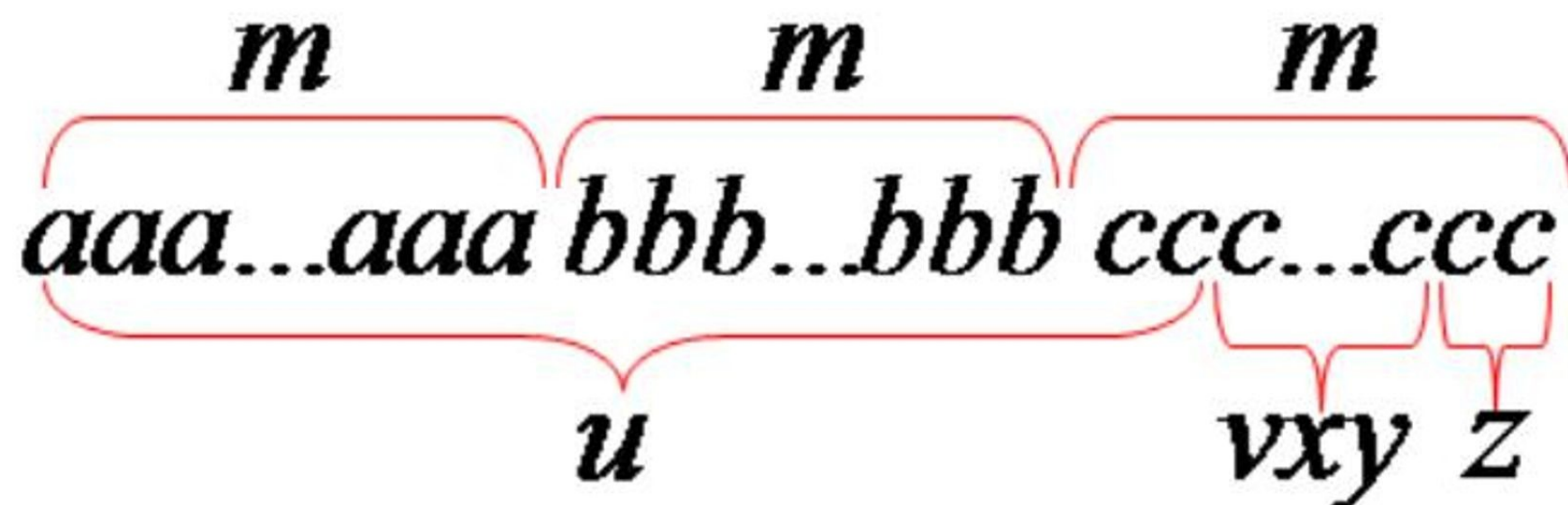
$$\begin{array}{c} \overbrace{aaa \dots aaa}^m \quad \overbrace{bbb \dots bbb}^m \quad \overbrace{ccc \dots ccc}^m \\ \underbrace{\hspace{1.5cm}}_u \quad \underbrace{\hspace{1.5cm}}_{vxy} \quad \underbrace{\hspace{1.5cm}}_z \end{array}$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 3: vxy is within c^m



$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 3: Similar analysis with case 1

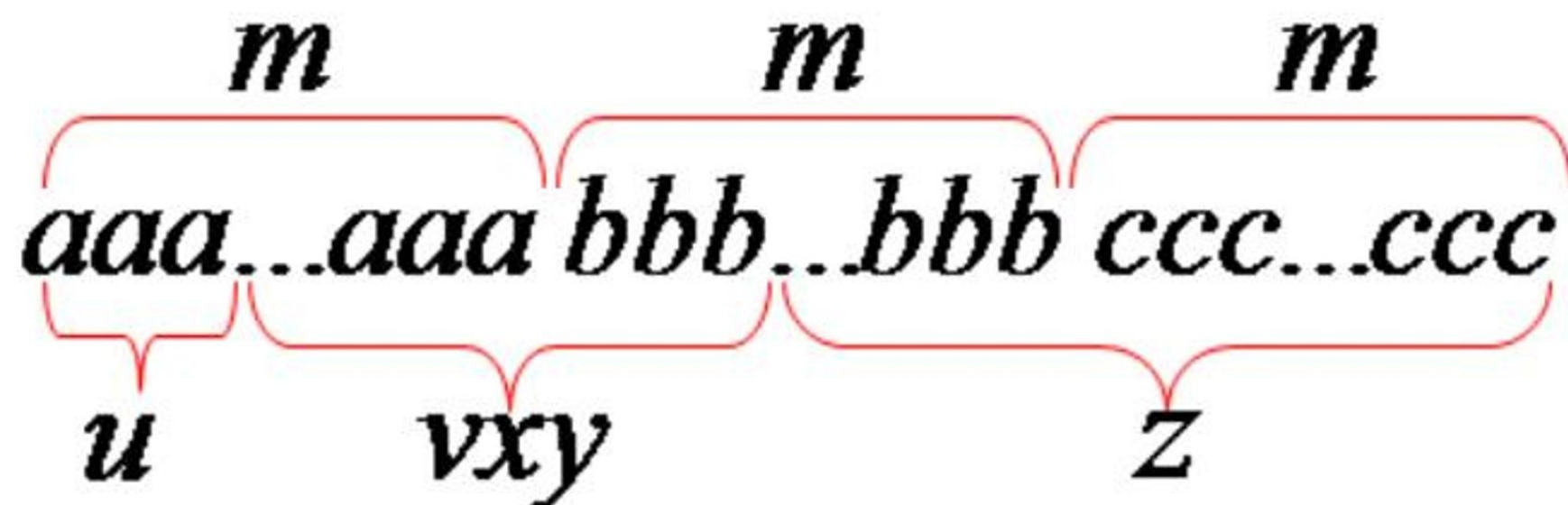
$$\begin{array}{c} m \qquad m \qquad m \\ \overbrace{aaa \dots aaa} \quad \overbrace{bbb \dots bbb} \quad \overbrace{ccc \dots ccc} \\ \underbrace{aaa \dots aaa \quad bbb \dots bbb}_{u} \quad \underbrace{cc}_{vxy} \quad c_z \end{array}$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 4: vxy overlaps a^m and b^m



$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 4: Possibility 1: v contains only a
 y contains only b

$$\overbrace{aaa \dots aaa}^m \overbrace{bbb \dots bbb}^m \overbrace{ccc \dots ccc}^m$$

$$\underbrace{\hspace{1.5cm}}_u \underbrace{\hspace{1.5cm}}_{vxy} \underbrace{\hspace{1.5cm}}_z$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 4: Possibility 1: v contains only a

$k_1 + k_2 \geq 1$ y contains only b

$$\begin{array}{c}
 \overbrace{aaa \dots aaaa}^{m+k_1} \overbrace{bbbbbb \dots bbb}^{m+k_2} \overbrace{ccc \dots ccc}^m \\
 \underbrace{aaa}_{u} \underbrace{aaaaaa}_{v^2} \underbrace{bbbbb}_{xy^2} \underbrace{ccc}_{z}
 \end{array}$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 4: From Pumping Lemma: $uv^2xy^2z \in L$

$$k_1 + k_2 \geq 1$$

$$\underbrace{\overbrace{aaa \dots a}^{m+k_1} \overbrace{bbb \dots b}^{m+k_2} \overbrace{ccc \dots c}^m}_{\substack{u \qquad v^2xy^2 \qquad z}}$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 4: From Pumping Lemma: $uv^2xy^2z \in L$

$$k_1 + k_2 \geq 1$$

However: $uv^2xy^2z = a^{m+k_1} b^{m+k_2} c^m \notin L$

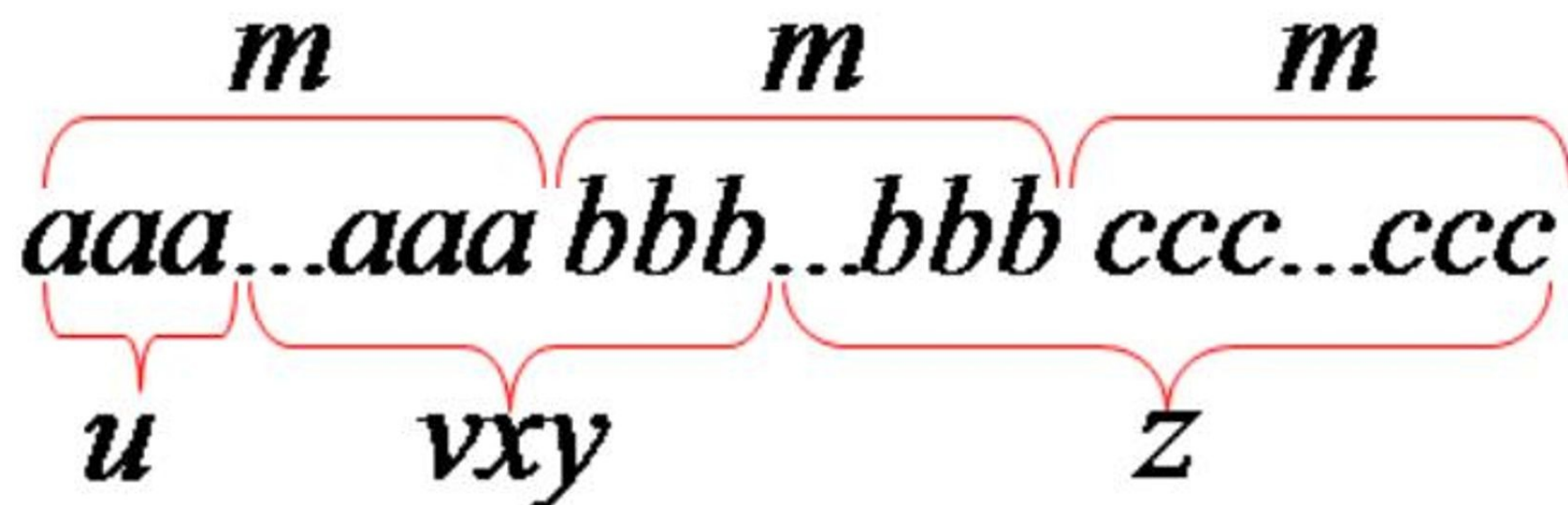
Contradiction!!!

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 4: Possibility 2: v contains a and b
 y contains only b



$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 4: Possibility 2: v contains a and b

$k_1 + k_2 + k \geq 1$ y contains only b

$$\begin{array}{ccccccc}
 & m & k_1 & k_2 & m+k & & m \\
 \underbrace{aaa \dots aaa}_{u} & \underbrace{aabb}_{v^2} & \underbrace{aabb}_{xy^2} & \underbrace{bbbbbb}_{xy^2} & \underbrace{bbb}_{xy^2} & \underbrace{ccc \dots ccc}_{z} & \\
 \end{array}$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 4: From Pumping Lemma: $uv^2xy^2z \in L$

$$k_1 + k_2 + k \geq 1$$

$$\underbrace{aaa \dots aaa}_{m} \underbrace{abb}_{k_1} \underbrace{aabb}_{k_2} \underbrace{bbbbbb \dots bbb}_{m+k} \underbrace{ccc \dots ccc}_m$$

$$\underbrace{u}_{aaa \dots aaa} \underbrace{v^2 xy^2}_{abb aabb bbb \dots bbb} \underbrace{z}_{ccc \dots ccc}$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 4: From Pumping Lemma: $uv^2xy^2z \in L$

However: $k_1 + k_2 + k \geq 1$

$$uv^2xy^2z = a^m b^{k_1} a^{k_2} b^{m+k} c^m \notin L$$

Contradiction!!!

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 4: Possibility 3: v contains only a
 y contains a and b

$$\begin{array}{c} \overbrace{aaa \dots aaa}^m \quad \overbrace{bbb \dots bbb}^m \quad \overbrace{ccc \dots ccc}^m \\ \underbrace{aaa}_{u} \quad \underbrace{aaa bbb}_{vxy} \quad \underbrace{bbb ccc}_{z} \end{array}$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 4: Possibility 3: v contains only a
 y contains a and b

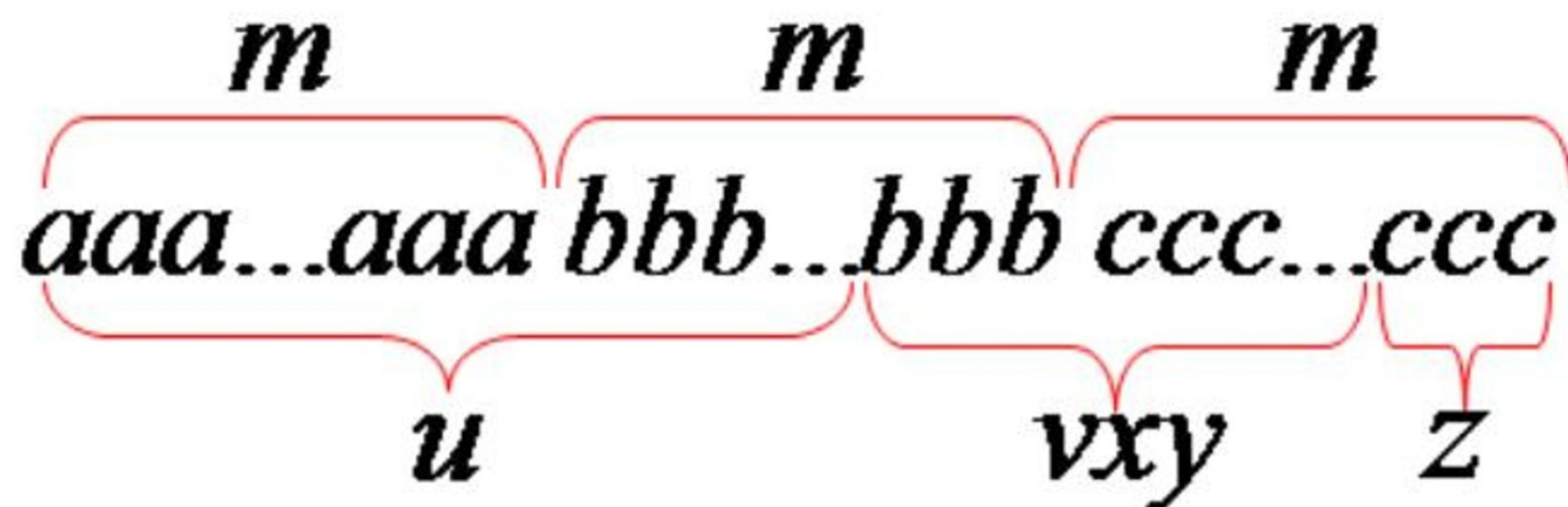
Similar analysis with Possibility 2

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 5: vxy overlaps b^m and c^m



$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 5: Similar analysis with case 4

$$\begin{array}{c} m \qquad m \qquad m \\ \overbrace{aaa \dots aaa} \quad \overbrace{bbb \dots bbb} \quad \overbrace{ccc \dots ccc} \\ \underbrace{\hspace{10em}}_u \quad \underbrace{\hspace{5em}}_{vxy} \quad \underbrace{\hspace{2em}}_z \end{array}$$

There are no other cases to consider

(since $|vxy| \leq m$, string vxy cannot

overlap a^m , b^m and c^m at the same time)

In all cases we obtained a contradiction

Therefore: The original assumption that
$$L = \{a^n b^n c^n : n \geq 0\}$$

is context-free must be wrong

Conclusion: L is not context-free